

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKENDDEVELOPMENT

Roll No: 2520030466

WEEK 2 : PRACTICAL- 1

```
Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\hp>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE skytrack_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE skytrack_db;
Database changed
mysql> CREATE TABLE Flights (
    -> flight_id INT PRIMARY KEY,
    -> flight_number VARCHAR(10) UNIQUE,
    -> source VARCHAR(50) NOT NULL,
    -> destination VARCHAR(50) NOT NULL,
    -> departure_date DATE,
    -> ticket_price DECIMAL(10,2),
    -> CHECK (ticket_price > 0)
    -> );
Query OK, 0 rows affected (0.11 sec)

mysql> CREATE TABLE Passengers (
    -> passenger_id INT PRIMARY KEY,
    -> passenger_name VARCHAR(100) NOT NULL,
    -> email VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE Bookings (
-> booking_id INT PRIMARY KEY,
-> passenger_id INT,
-> flight_id INT,
-> booking_date DATE,
-> FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),
-> FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> SHOW TABLES;
```

```
+-----+
| Tables_in_skytrack_db |
+-----+
| bookings               |
| flights                |
| passengers              |
+-----+
```

3 rows in set (0.00 sec)

```
mysql> DESCRIBE Flights;
```

Field	Type	Null	Key	Default	Extra
flight_id	int	NO	PRI	NULL	
flight_number	varchar(10)	YES	UNI	NULL	
source	varchar(50)	NO		NULL	
destination	varchar(50)	NO		NULL	
departure_date	date	YES		NULL	
ticket_price	decimal(10,2)	YES		NULL	

6 rows in set (0.00 sec)

```
mysql> DESCRIBE Passengers;
```

Field	Type	Null	Key	Default	Extra
passenger_id	int	NO	PRI	NULL	
passenger_name	varchar(100)	NO		NULL	
email	varchar(100)	YES	UNI	NULL	

```
3 rows in set (0.00 sec)
```

```
mysql> DESCRIBE Bookings;
```

Field	Type	Null	Key	Default	Extra
booking_id	int	NO	PRI	NULL	
passenger_id	int	YES	MUL	NULL	
flight_id	int	YES	MUL	NULL	
booking_date	date	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Flights VALUES
```

```
-> (1, 'AI101', 'Hyderabad', 'Delhi', '2026-08-01', 5500),  
-> (2, 'AI102', 'Delhi', 'Mumbai', '2026-08-02', 4500),  
-> (3, 'AI103', 'Mumbai', 'Chennai', '2026-08-03', 4000),  
-> (4, 'AI104', 'Chennai', 'Bangalore', '2026-08-04', 3000),  
-> (5, 'AI105', 'Bangalore', 'Hyderabad', '2026-08-05', 3500),  
-> (6, 'AI106', 'Hyderabad', 'Pune', '2026-08-06', 4200),  
-> (7, 'AI107', 'Pune', 'Kolkata', '2026-08-07', 6000),  
-> (8, 'AI108', 'Kolkata', 'Delhi', '2026-08-08', 5200),  
-> (9, 'AI109', 'Delhi', 'Goa', '2026-08-09', 4800),  
-> (10, 'AI110', 'Goa', 'Hyderabad', '2026-08-10', 5000);
```

```
Query OK, 10 rows affected (0.01 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Passengers VALUES
```

```
-> (1, 'Rahul', 'rahul@gmail.com'),  
-> (2, 'Sneha', 'sneha@gmail.com'),  
-> (3, 'Kiran', 'kiran@gmail.com'),  
-> (4, 'Anjali', 'anjali@gmail.com'),  
-> (5, 'Vikram', 'vikram@gmail.com'),  
-> (6, 'Priya', 'priya@gmail.com'),  
-> (7, 'Arjun', 'arjun@gmail.com'),  
-> (8, 'Meena', 'meena@gmail.com'),  
-> (9, 'Ravi', 'ravi@gmail.com'),  
-> (10, 'Sita', 'sita@gmail.com');
```

```
Query OK, 10 rows affected (0.01 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Bookings VALUES
```

```
-> (101, 1, 1, '2026-07-20'),  
-> (102, 2, 2, '2026-07-21'),  
-> (103, 3, 3, '2026-07-22'),  
-> (104, 4, 4, '2026-07-23'),  
-> (105, 5, 5, '2026-07-24'),  
-> (106, 6, 6, '2026-07-25'),  
-> (107, 7, 7, '2026-07-26'),  
-> (108, 8, 8, '2026-07-27'),  
-> (109, 9, 9, '2026-07-28'),  
-> (110, 10, 10, '2026-07-29');
```

```
Query OK, 10 rows affected (0.01 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Flights;
```

flight_id	flight_number	source	destination	departure_date	ticket_price
1	AI101	Hyderabad	Delhi	2026-08-01	5500.00
2	AI102	Delhi	Mumbai	2026-08-02	4500.00
3	AI103	Mumbai	Chennai	2026-08-03	4000.00
4	AI104	Chennai	Bangalore	2026-08-04	3000.00
5	AI105	Bangalore	Hyderabad	2026-08-05	3500.00
6	AI106	Hyderabad	Pune	2026-08-06	4200.00
7	AI107	Pune	Kolkata	2026-08-07	6000.00
8	AI108	Kolkata	Delhi	2026-08-08	5200.00
9	AI109	Delhi	Goa	2026-08-09	4800.00
10	AI110	Goa	Hyderabad	2026-08-10	5000.00

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Passengers;
```

passenger_id	passenger_name	email
1	Rahul	rahul@gmail.com
2	Sneha	sneha@gmail.com
3	Kiran	kiran@gmail.com
4	Anjali	anjali@gmail.com
5	Vikram	vikram@gmail.com
6	Priya	priya@gmail.com
7	Arjun	arjun@gmail.com
8	Meena	meena@gmail.com
9	Ravi	ravi@gmail.com
10	Sita	sita@gmail.com

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Bookings;
```

booking_id	passenger_id	flight_id	booking_date
101	1	1	2026-07-20
102	2	2	2026-07-21
103	3	3	2026-07-22
104	4	4	2026-07-23
105	5	5	2026-07-24
106	6	6	2026-07-25
107	7	7	2026-07-26
108	8	8	2026-07-27
109	9	9	2026-07-28
110	10	10	2026-07-29

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT
-> Passengers.passenger_name,
-> Flights.flight_number,
-> Flights.source,
-> Flights.destination
-> FROM Bookings
-> INNER JOIN Passengers
-> ON Bookings.passenger_id = Passengers.passenger_id
-> INNER JOIN Flights
-> ON Bookings.flight_id = Flights.flight_id;
```

```
mysql> SELECT
-> Passengers.passenger_name,
-> Flights.flight_number,
-> Flights.source,
-> Flights.destination
-> FROM Bookings
-> INNER JOIN Passengers
-> ON Bookings.passenger_id = Passengers.passenger_id
-> INNER JOIN Flights
-> ON Bookings.flight_id = Flights.flight_id;
```

passenger_name	flight_number	source	destination
Rahul	AI101	Hyderabad	Delhi
Sneha	AI102	Delhi	Mumbai
Kiran	AI103	Mumbai	Chennai
Anjali	AI104	Chennai	Bangalore
Vikram	AI105	Bangalore	Hyderabad
Priya	AI106	Hyderabad	Pune
Arjun	AI107	Pune	Kolkata
Meena	AI108	Kolkata	Delhi
Ravi	AI109	Delhi	Goa
Sita	AI110	Goa	Hyderabad

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT
-> destination,
-> COUNT(*) AS total_flights
-> FROM Flights
-> GROUP BY destination;
```

destination	total_flights
Delhi	2
Mumbai	1
Chennai	1
Bangalore	1
Hyderabad	2
Pune	1
Kolkata	1
Goa	1

8 rows in set (0.01 sec)

```
mysql> CREATE TABLE Flight_History(
-> history_id INT PRIMARY KEY,
-> flight_id INT,
-> flight_number VARCHAR(10),
-> action_date DATE
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> INSERT INTO Flights
```

```
-> VALUES
```

```
-> (11, 'AI111', 'Hyderabad', 'Jaipur', '2026-08-11', 6200);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> INSERT INTO Flight_History
```

```
-> VALUES
```

```
-> (1, 11, 'AI111', '2026-07-30');
```

Query OK, 1 row affected (0.00 sec)

```
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> CREATE INDEX idx_flight_number
-> ON Flights(flight_number);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Flights
-> WHERE flight_number='AI105';
```

flight_id	flight_number	source	destination	departure_date	ticket_price
5	AI105	Bangalore	Hyderabad	2026-08-05	3500.00

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM Flights
-> WHERE destination='Hyderabad';
```

flight_id	flight_number	source	destination	departure_date	ticket_price
5	AI105	Bangalore	Hyderabad	2026-08-05	3500.00
10	AI110	Goa	Hyderabad	2026-08-10	5000.00

```
2 rows in set (0.00 sec)
```

Week 2 Practical 2:

```
Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\hp>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> ^C
mysql> CREATE DATABASE medicare_db;
ERROR 1007 (HY000): Can't create database 'medicare_db'; database exists
mysql> CREATE DATABASE medicare_db;
ERROR 1007 (HY000): Can't create database 'medicare_db'; database exists
mysql> clear
mysql> USE medicare_db;
Database changed
mysql> CREATE TABLE Doctors (
    -> doctor_id INT PRIMARY KEY,
    -> doctor_name VARCHAR(100) NOT NULL,
    -> specialization VARCHAR(100),
    -> consultation_fee DECIMAL(10,2),
    -> CHECK (consultation_fee > 0)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE Patients (
    -> patient_id INT PRIMARY KEY,
    -> patient_name VARCHAR(100) NOT NULL,
    -> email VARCHAR(100) UNIQUE
    -> );
```

```
-> );  
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> CREATE TABLE Appointments (  
-> appointment_id INT PRIMARY KEY,  
-> doctor_id INT,  
-> patient_id INT,  
-> appointment_date DATE,  
-> FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),  
-> FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)  
-> );  
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> SHOW TABLES;  
+-----+  
| Tables_in_medicare_db |  
+-----+  
| appointments          |  
| doctors               |  
| patients              |  
+-----+  
3 rows in set (0.02 sec)
```

```
mysql> DESCRIBE Doctors;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| doctor_id      | int           | NO   | PRI | NULL    |       |  
| doctor_name    | varchar(100)  | NO   |     | NULL    |       |  
| specialization | varchar(100)  | YES  |     | NULL    |       |  
| consultation_fee | decimal(10,2) | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.01 sec)
```

```
mysql> DESCRIBE Patients;
```

Field	Type	Null	Key	Default	Extra
patient_id	int	NO	PRI	NULL	
patient_name	varchar(100)	NO		NULL	
email	varchar(100)	YES	UNI	NULL	

```
3 rows in set (0.00 sec)
```

```
mysql> DESCRIBE Appointments;
```

Field	Type	Null	Key	Default	Extra
appointment_id	int	NO	PRI	NULL	
doctor_id	int	YES	MUL	NULL	
patient_id	int	YES	MUL	NULL	
appointment_date	date	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Doctors VALUES
```

```
-> (1,'Dr. Ramesh','Cardiology',800),  
-> (2,'Dr. Suresh','Neurology',900),  
-> (3,'Dr. Kavitha','Orthopedics',700),  
-> (4,'Dr. Anjali','Dermatology',600),  
-> (5,'Dr. Rajesh','Pediatrics',500),  
-> (6,'Dr. Mahesh','ENT',550),  
-> (7,'Dr. Priya','Gynecology',750),  
-> (8,'Dr. Arjun','Ophthalmology',650),  
-> (9,'Dr. Kiran','Cardiology',850),  
-> (10,'Dr. Meena','General Medicine',400);
```

```
Query OK, 10 rows affected (0.05 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Patients VALUES
```

```
-> (1,'Rahul','rahul@gmail.com'),  
-> (2,'Sneha','sneha@gmail.com'),  
-> (3,'Kiran','kiran@gmail.com'),  
-> (4,'Anjali','anjali@gmail.com'),  
-> (5,'Vikram','vikram@gmail.com'),
```

```
-> (1,'Dr. Ramesh','Cardiology',800),
-> (2,'Dr. Suresh','Neurology',900),
-> (3,'Dr. Kavitha','Orthopedics',700),
-> (4,'Dr. Anjali','Dermatology',600),
-> (5,'Dr. Rajesh','Pediatrics',500),
-> (6,'Dr. Mahesh','ENT',550),
-> (7,'Dr. Priya','Gynecology',750),
-> (8,'Dr. Arjun','Ophthalmology',650),
-> (9,'Dr. Kiran','Cardiology',850),
-> (10,'Dr. Meena','General Medicine',400);
Query OK, 10 rows affected (0.05 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Patients VALUES
-> (1,'Rahul','rahul@gmail.com'),
-> (2,'Sneha','sneha@gmail.com'),
-> (3,'Kiran','kiran@gmail.com'),
-> (4,'Anjali','anjali@gmail.com'),
-> (5,'Vikram','vikram@gmail.com'),
-> (6,'Priya','priya@gmail.com'),
-> (7,'Arjun','arjun@gmail.com'),
-> (8,'Meena','meena@gmail.com'),
-> (9,'Ravi','ravi@gmail.com'),
-> (10,'Sita','sita@gmail.com');
Query OK, 10 rows affected (0.04 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Appointments VALUES
-> (101,1,1,'2026-08-01'),
-> (102,2,2,'2026-08-02'),
-> (103,3,3,'2026-08-03'),
-> (104,4,4,'2026-08-04'),
-> (105,5,5,'2026-08-05'),
-> (106,6,6,'2026-08-06'),
-> (107,7,7,'2026-08-07'),
-> (108,8,8,'2026-08-08'),
-> (109,9,9,'2026-08-09'),
-> (110,10,10,'2026-08-10');
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Doctors;
```

doctor_id	doctor_name	specialization	consultation_fee
1	Dr. Ramesh	Cardiology	800.00
2	Dr. Suresh	Neurology	900.00
3	Dr. Kavitha	Orthopedics	700.00
4	Dr. Anjali	Dermatology	600.00
5	Dr. Rajesh	Pediatrics	500.00
6	Dr. Mahesh	ENT	550.00
7	Dr. Priya	Gynecology	750.00
8	Dr. Arjun	Ophthalmology	650.00
9	Dr. Kiran	Cardiology	850.00
10	Dr. Meena	General Medicine	400.00

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Patients;
```

patient_id	patient_name	email
1	Rahul	rahul@gmail.com
2	Sneha	sneha@gmail.com
3	Kiran	kiran@gmail.com
4	Anjali	anjali@gmail.com
5	Vikram	vikram@gmail.com
6	Priya	priya@gmail.com
7	Arjun	arjun@gmail.com
8	Meena	meena@gmail.com
9	Ravi	ravi@gmail.com
10	Sita	sita@gmail.com

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Appointments;
```

appointment_id	doctor_id	patient_id	appointment_date
101	1	1	2026-08-01
102	2	2	2026-08-02
103	3	3	2026-08-03
104	4	4	2026-08-04
105	5	5	2026-08-05
106	6	6	2026-08-06
107	7	7	2026-08-07
108	8	8	2026-08-08
109	9	9	2026-08-09
110	10	10	2026-08-10

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT
```

```
-> Patients.patient_name,  
-> Doctors.doctor_name,  
-> Doctors.specialization,  
-> Appointments.appointment_date  
-> FROM Appointments  
-> INNER JOIN Patients  
-> ON Appointments.patient_id = Patients.patient_id  
-> INNER JOIN Doctors  
-> ON Appointments.doctor_id = Doctors.doctor_id;
```

patient_name	doctor_name	specialization	appointment_date
Rahul	Dr. Ramesh	Cardiology	2026-08-01
Sneha	Dr. Suresh	Neurology	2026-08-02
Kiran	Dr. Kavitha	Orthopedics	2026-08-03
Anjali	Dr. Anjali	Dermatology	2026-08-04
Vikram	Dr. Rajesh	Pediatrics	2026-08-05
Priya	Dr. Mahesh	ENT	2026-08-06
Arjun	Dr. Priya	Gynecology	2026-08-07
Meena	Dr. Arjun	Ophthalmology	2026-08-08
Ravi	Dr. Kiran	Cardiology	2026-08-09
Sita	Dr. Meena	General Medicine	2026-08-10

Arjun	Dr. Priya	Gynecology	2026-08-07	
Meena	Dr. Arjun	Ophthalmology	2026-08-08	
Ravi	Dr. Kiran	Cardiology	2026-08-09	
Sita	Dr. Meena	General Medicine	2026-08-10	

+-----+-----+-----+-----+
10 rows in set (0.00 sec)

```
mysql> SELECT
-> specialization,
-> COUNT(*) AS total_doctors
-> FROM Doctors
-> GROUP BY specialization;
```

specialization	total_doctors	
Cardiology	2	
Neurology	1	
Orthopedics	1	
Dermatology	1	
Pediatrics	1	
ENT	1	
Gynecology	1	
Ophthalmology	1	
General Medicine	1	

+-----+-----+-----+-----+
9 rows in set (0.01 sec)

```
mysql> CREATE TABLE Doctor_History (
-> history_id INT PRIMARY KEY,
-> doctor_id INT,
-> doctor_name VARCHAR(100),
-> registration_date DATE
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Doctors
    -> VALUES
    -> (11,'Dr. Sanjay','Cardiology',900);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Doctor_History
    -> VALUES
    -> (1,11,'Dr. Sanjay','2026-07-30');
Query OK, 1 row affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_specialization
    -> ON Doctors(specialization);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Doctors
    -> WHERE specialization='Cardiology';
+-----+-----+-----+-----+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----+-----+-----+-----+
|          1 | Dr. Ramesh  | Cardiology     |          800.00 |
|          9 | Dr. Kiran   | Cardiology     |          850.00 |
|         11 | Dr. Sanjay  | Cardiology     |          900.00 |
+-----+-----+-----+-----+
3 rows in set (0.01 sec)

```